Continuous Random Variables

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Verifying a valid probability density function**

A continuous random variable X has probability density function $f(x) = \frac{1}{4}$ for $0 \le x \le 4$, and $f(x) = 0$ otherwise. Verify that this is a valid pdf.

Since $f(x) = \frac{1}{4} \ge 0$ throughout, and $\int_0^4 \frac{1}{4}\,dx = \frac{1}{4}(4) = 1$, the total area under f equals 1, confirming a valid pdf (the uniform distribution on $[0, 4]$).

2 **[Foundational]** • **Target: Verifying a valid probability density function**

Determine the value of c such that $f(x) = c$ for $2 \le x \le 6$, and $f(x) = 0$ otherwise, is a valid pdf.

$\int_2^6 c\,dx = 4c = 1 \Rightarrow c = \frac{1}{4}$.

3 **[Foundational]** • **Target: Finding an unknown constant in a pdf**

A continuous random variable X has pdf $f(x) = kx$ for $0 \le x \le 5$, and $f(x) = 0$ otherwise. Find k.

$\int_0^5 kx\,dx = k\left[\frac{x^2}{2}\right]_0^5 = \frac{25k}{2} = 1 \Rightarrow k = \frac{2}{25}$.

4 **[Foundational]** • **Target: Finding probabilities by integrating the pdf**

Using the same pdf, find $P(2 \le X \le 4)$.

$\int_2^4 \frac{2}{25}x\,dx = \frac{2}{25}\left[\frac{x^2}{2}\right]_2^4 = \frac{2}{25}(8 - 2) = \frac{12}{25} = 0.48$.

5 **[Foundational]** • **Target: The cumulative distribution function**

For the pdf $f(x) = \frac{2}{25}x$ on $0 \le x \le 5$, find the cumulative distribution function $F(x)$.

$F(x) = \int_0^x \frac{2}{25}t\,dt = \frac{2}{25} \cdot \frac{x^2}{2} = \frac{x^2}{25}$, for $0 \le x \le 5$.

6 **[Foundational]** • **Target: The cumulative distribution function**

Using $F(x) = \frac{x^2}{25}$, find $P(X \le 3)$ and hence $P(X > 3)$.

$F(3) = \frac{9}{25} = 0.36$. $P(X > 3) = 1 - 0.36 = 0.64$.

7 **[Foundational]** • **Target: Expectation of a continuous random variable**

Using $f(x) = \frac{2}{25}x$ for $0 \le x \le 5$, find $E(X)$, using $E(X) = \int x f(x)\,dx$.

$E(X) = \int_0^5 x \cdot \frac{2}{25}x\,dx = \frac{2}{25}\left[\frac{x^3}{3}\right]_0^5 = \frac{2}{25} \cdot \frac{125}{3} = \frac{10}{3} \approx 3.333$.

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Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 **[Intermediate]** • **Target: Expectation of a continuous random variable**

Using the uniform pdf $f(x) = \frac{1}{4}$ for $0 \le x \le 4$, find $E(X)$, and verify it matches the midpoint of the interval.

$E(X) = \int_0^4 x \cdot \frac{1}{4}\,dx = \frac{1}{4}\left[\frac{x^2}{2}\right]_0^4 = \frac{1}{4}(8) = 2$, matching the midpoint of $[0, 4]$, as expected for a uniform distribution.

9 **[Intermediate]** • **Target: Expectation of a continuous random variable**

Using $f(x) = \frac{1}{9}x^2$ on $[0, 3]$, find $E(X)$.

$E(X) = \int_0^3 x \cdot \frac{1}{9}x^2\,dx = \frac{1}{9}\left[\frac{x^4}{4}\right]_0^3 = \frac{1}{9} \cdot \frac{81}{4} = \frac{9}{4} = 2.25.$

10 **[Intermediate]** • **Target: Variance of a continuous random variable**

Using $f(x) = \frac{2}{25}x$ on $[0, 5]$ and $E(X) = \frac{10}{3}$ (found earlier), find $\mathrm{Var}(X)$, using $\mathrm{Var}(X) = E(X^2) - [E(X)]^2$.

$E(X^2) = \int_0^5 x^2 \cdot \frac{2}{25}x\,dx = \frac{2}{25}\left[\frac{x^4}{4}\right]_0^5 = \frac{2}{25} \cdot \frac{625}{4} = 12.5.$ $\mathrm{Var}(X) = 12.5 - \left(\frac{10}{3}\right)^2 = 12.5 - \frac{100}{9} \approx 1.389.$

11 **[Intermediate]** • **Target: Variance of a continuous random variable**

Using the uniform pdf $f(x) = \frac{1}{4}$ on $[0, 4]$ and $E(X) = 2$ (found earlier), find $\mathrm{Var}(X)$.

$E(X^2) = \int_0^4 x^2 \cdot \frac{1}{4}\,dx = \frac{1}{4}\left[\frac{x^3}{3}\right]_0^4 = \frac{1}{4} \cdot \frac{64}{3} = \frac{16}{3}.$ $\mathrm{Var}(X) = \frac{16}{3} - 2^2 = \frac{16}{3} - \frac{12}{3} = \frac{4}{3} \approx 1.333.$

12 **[Intermediate]** • **Target: Variance of a continuous random variable**

Using $f(x) = \frac{1}{9}x^2$ on $[0, 3]$ and $E(X) = \frac{9}{4}$ (found earlier), find $\mathrm{Var}(X)$.

$E(X^2) = \int_0^3 x^2 \cdot \frac{1}{9}x^2\,dx = \frac{1}{9}\left[\frac{x^5}{5}\right]_0^3 = \frac{1}{9} \cdot \frac{243}{5} = \frac{27}{5} = 5.4.$ $\mathrm{Var}(X) = 5.4 - \left(\frac{9}{4}\right)^2 = 5.4 - 5.0625 = 0.3375.$

13 **[Intermediate]** • **Target: Median and quartiles**

Using the cdf $F(x) = \frac{x^2}{25}$ for $0 \le x \le 5$, find the median m of X, solving $F(m) = 0.5$.

$\frac{m^2}{25} = 0.5 \Rightarrow m^2 = 12.5 \Rightarrow m = \sqrt{12.5} \approx 3.536.$

14 **[Intermediate]** • **Target: Median and quartiles**

Using the same cdf, find the lower quartile Q_1, solving $F(Q_1) = 0.25$.

$\frac{Q_1^2}{25} = 0.25 \Rightarrow Q_1^2 = 6.25 \Rightarrow Q_1 = 2.5.$

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

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15 **[Challenging]** • **Target: Median and quartiles** • **Cross-Lesson: 14.01 Sampling & Estimation**

Using $F(x) = \frac{x}{2} - \frac{x^2}{16}$ for $0 \le x \le 4$, find the upper quartile Q_3, solving $F(Q_3) = 0.75$.

$\frac{x}{2} - \frac{x^2}{16} = 0.75$. Multiplying by 16: $8x - x^2 = 12 \Rightarrow x^2 - 8x + 12 = 0 \Rightarrow (x-2)(x-6) = 0$. Since the domain is $[0, 4]$, $x = 6$ is rejected, so $Q_3 = 2$.

16 **[Challenging]** • **Target: Median and quartiles** • **Cross-Lesson: 15.01 Hypothesis Testing**

Using the cdf $F(x) = \frac{x^3}{27}$ for $0 \le x \le 3$, find the median m of X, to three decimal places.

$\frac{m^3}{27} = 0.5 \Rightarrow m^3 = 13.5 \Rightarrow m = 13.5^{1/3} \approx 2.381$.

17 **[Challenging]** • **Target: A well-designed word problem: waiting time at a bus stop** • **Cross-Lesson: 2.01 Exponentials & Logarithms**

This question has four parts. The waiting time X (minutes) for a bus is modeled by $f(x) = \frac{3}{64}x^2$ for $0 \le x \le 4$, and $f(x) = 0$ otherwise. (a) Verify this is a valid pdf. (b) Find the cdf $F(x)$, and use it to find the probability a passenger waits more than 3 minutes. (c) Find $E(X)$, the mean waiting time. (d) Find the median waiting time, to three decimal places.

(a) $\int_0^4 \frac{3}{64}x^2\,dx = \frac{3}{64}\left[\frac{x^3}{3}\right]_0^4 = \frac{3}{64} \cdot \frac{64}{3} = 1$, and $f(x) \ge 0$ throughout, confirming a valid pdf. (b) $F(x) = \int_0^x \frac{3}{64}t^2\,dt = \frac{x^3}{64}$. $P(X > 3) = 1 - F(3) = 1 - \frac{27}{64} = \frac{37}{64} \approx 0.578$. (c) $E(X) = \int_0^4 x \cdot \frac{3}{64}x^2\,dx = \frac{3}{64}\left[\frac{x^4}{4}\right]_0^4 = \frac{3}{64}(64) = 3$ minutes. (d) Solving $F(m) = 0.5$: $\frac{m^3}{64} = 0.5 \Rightarrow m^3 = 32 \Rightarrow m = 32^{1/3} \approx 3.175$ minutes.

18 **[Challenging]** • **Target: A well-designed word problem: waiting time at a bus stop** • **Cross-Lesson: 3.01 Advanced Trigonometry**

A continuous random variable X has probability density function $f(x) = kx(2 - x)$ for $0 \le x \le 2$ and $f(x) = 0$ elsewhere. Determine: (a) the constant k; (b) the median of X; (c) the variance $\mathrm{Var}(X)$.

1. Normalization condition: $\int_0^2 k(2x - x^2)\,dx = 1 \Rightarrow k\left[x^2 - \frac{x^3}{3}\right]_0^2 = k\left(4 - \frac{8}{3}\right) = k\left(\frac{4}{3}\right) = 1 \Rightarrow k = \frac{3}{4}$. 2. Median m: By visual symmetry of the parabola $f(x) = \frac{3}{4}x(2 - x)$ about $x = 1$, $m = 1$. We verify: $F(1) = \int_0^1 \frac{3}{4}(2x - x^2)\,dx = \frac{3}{4}\left[x^2 - \frac{x^3}{3}\right]_0^1 = \frac{3}{4}\left(\frac{2}{3}\right) = \frac{1}{2}$. ✓ 3. Variance: By symmetry, $E[X] = 1$..

19 **[Challenging]** • **Target: Verifying a valid probability density function** • **Cross-Lesson: 4.01 Differentiation**

A continuous random variable X has probability density function $f(x) = 2e^{-2x}$ for $x \ge 0$ and 0 elsewhere. Find the cumulative distribution function $F(x)$ and determine the exact median of X.

1. Find CDF for $x \ge 0$: $F(x) = \int_0^x 2e^{-2t}\,dt = [-e^{-2t}]_0^x = 1 - e^{-2x}$. 2. Set $F(m) = 0.5$ to find the median m: $1 - e^{-2m} = 0.5 \Rightarrow e^{-2m} = 0.5 \Rightarrow -2m = \ln(0.5) = -\ln 2 \Rightarrow m = \frac{\ln 2}{2}$.

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20 [Challenging] • Target: Finding an unknown constant in a pdf • Cross-Lesson: 5.01 Integration

A continuous random variable X has PDF given by $f(x) = \frac{k}{x^2}$ for $1 \le x \le 4$, and 0 elsewhere. (a) Show that $k = \frac{4}{3}$. (b) Find the median of X.

(a) Integrate over the support and equate to 1: $\int_1^4 \frac{k}{x^2}\,dx = k[-\frac{1}{x}]_1^4 = k(-\frac{1}{4} - (-1)) = \frac{3}{4}k = 1 \Rightarrow k = \frac{4}{3}$ (b) Find the CDF for $1 \le x \le 4$: $F(x) = \int_1^x \frac{4}{3t^2}\,dt = \frac{4}{3}(1 - \frac{1}{x})$ Set $F(m) = 0.5$ to find the median m.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Verifying a valid probability density function

A discrete random variable, X, can take the values 1, 2, 3, 4, and 5. The probability distribution of X is given by $P(X = x) = kx$ for $x \in \{1, 2, 3, 4, 5\}$, where k is a constant. Find the value of k.

a. $\frac{1}{15}$ ✓

b. $\frac{1}{5}$

c. $\frac{1}{3}$

d. 1

The sum of probabilities for all possible values of a discrete random variable must equal 1. Therefore, $\sum_{x=1}^{5} P(X = x) = 1$. Given $P(X = x) = kx$, we have: $P(X = 1) = k(1) = k$ $P(X = 2) = k(2) = 2k$ $P(X = 3) = k(3) = 3k$ $P(X = 4) = k(4) = 4k$ $P(X = 5) = k(5) = 5k$ Summing these probabilities: $k + 2k + 3k + 4k + 5k = 1$ $15k = 1$ $k = \frac{1}{15}$ Therefore, the value of k is $\frac{1}{15}$.

22 [Foundational] • Target: Finding an unknown constant in a pdf

A discrete random variable X has probability distribution given by $P(X = 1) = k, P(X = 2) = k, P(X = 3) = 2k$, where k is a constant. Find k.

a. 1/3

b. **1/4** ✓

c. 1/5

d. 1/2

Since probabilities sum to 1: $(1 + 1 + 2)k = 1 \Rightarrow 4k = 1 \Rightarrow k = 1/4$

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23 [Foundational] • Target: Finding probabilities by integrating the pdf

The rational function f is defined for all real $xotin\{-1, 2\}$ by $f(x) = \frac{x^3 + kx + 7}{(x+1)(x-2)^2}$ where k is a constant. The partial fraction decomposition of $f(x)$ takes the form $A + \frac{B}{x+1} + \frac{C}{x-2} + \frac{D}{(x-2)^2}$ where $A, B, C,$ and D are real constants. Given that $C = 2$, find the value of k.

a. -3 ✓

b. 1

c. 3

d. 24

Step 1: Determine the polynomial quotient A. The degree of the numerator is 3 and the degree of the denominator is $1 + 2 = 3$. Expanding the denominator gives:
$(x+1)(x-2)^2 = (x+1)(x^2 - 4x + 4) = x^3 - 3x^2 + 4$ Since the leading coefficient of both the numerator and the denominator is 1, polynomial division yields a constant quotient: $A = \frac{1}{1} = 1$ Step 2: Set up the algebraic identity. Multiplying both sides of the partial fraction expansion by $(x+1)(x-2)^2$.

24 [Foundational] • Target: The cumulative distribution function

A discrete random variable X has distribution $P(X = 1) = 2/5, P(X = 2) = 1/5, P(X = 3) = 2/5$. Given $Y = 5X - 3$, find $E(Y)$.

a. 6

b. **7** ✓

c. 8

d. 13

Applying Discrete Random Variables principles, $E(X) = 2$. $E(Y) = aE(X) + b = 5(2) - 3 = 7$ Therefore, the verified correct option is 7.

25 [Foundational] • Target: Using the cdf to find probabilities

A discrete random variable X takes values $0, 3, 4$ with probabilities $\frac{1}{10}, \frac{3}{10}, \frac{3}{5}$ respectively. Find $E(X)$.

a. 1

b. 6

c. $\frac{33}{10}$ ✓

d. 2

Working through this statistics step: $E(X) = \sum xP(X = x) = 0(\frac{1}{10}) + 3(\frac{3}{10}) + 4(\frac{3}{5}) = \frac{33}{10}$

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26 [Foundational] • Target: Expectation of a continuous random variable

A discrete random variable X has distribution $P(X=1)=1/10, P(X=2)=2/5, P(X=3)=1/2$. Find $\text{Var}(X)$.

a. 9/13

b. **11/25** ✓

c. 26/37

d. 9/10

$E(X)=12/5, E(X^2)=31/5$. $\text{Var}(X)=E(X^2)-[E(X)]^2=31/5-(12/5)^2=11/25$

27 [Foundational] • Target: Variance of a continuous random variable

A discrete random variable X has distribution $P(X=1)=2/5, P(X=2)=3/10, P(X=3)=3/10$. State $P(X=2)$.

a. 3/4

b. 8/9

c. **3/10** ✓

d. 11/15

Reading directly from the distribution table, $P(X=2)=3/10$.

28 [Foundational] • Target: Median and quartiles

A random variable $X \sim B(51, 0.4)$ is approximated by a normal distribution. State the variance of the approximating normal distribution.

a. **12.24** ✓

b. 15.24

c. 9.24

d. 19.24

Variance $= np(1-p) = 51 \times 0.4 \times 0.6 \approx 12.24$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: A well-designed word problem: waiting time at a bus stop

The rational function $f(x)$ is defined by $f(x) = \frac{5x+7}{(x-1)(x+3)}, \quad xeq1,\ xeq-3.$ Which of the following gives $f(x)$ expressed in partial fractions?

a. $\frac{3}{x-1} + \frac{2}{x+3}$ ✓

b. $\frac{2}{x-1} + \frac{3}{x+3}$

c. $\frac{3}{x-1} - \frac{2}{x+3}$

d. $\frac{4}{x-1} + \frac{1}{x+3}$

To decompose $f(x)$ into partial fractions, set up the identity: $\frac{5x+7}{(x-1)(x+3)} \equiv \frac{A}{x-1} + \frac{B}{x+3}$ Multiplying through by the common denominator $(x-1)(x+3)$: $5x+7 = A(x+3) + B(x-1)$ To find A, substitute $x = 1$: $5(1)+7 = A(1+3) + B(0) \Rightarrow 12 = 4A \Rightarrow A = 3$ To find B, substitute $x = -3$: $5(-3)+7 = A(0) + B(-3-1) \Rightarrow -8 = -4B \Rightarrow B = 2$ Therefore, the partial fraction decomposition is.

30 [Intermediate] • Target: Verifying a valid probability density function

Formulate the null and alternative hypotheses for the following scenario: "A company claims its average delivery time is 3 days. A customer believes it takes longer."

a. $H_0 : \mu = 10,\ H_1 : \mu < 10$

b. $H_0 : \mu = 70,\ H_1 : \mu > 70$

c. $H_0 : \mu = 500,\ H_1 : \mu < 500$

d. $H_0 : \mu = 3,\ H_1 : \mu > 3$ ✓

Applying Hypothesis Testing principles, $H_0 : \mu = 3,\ H_1 : \mu > 3$. Therefore, the verified correct option is $H_0 : \mu = 3,\ H_1 : \mu > 3$.

31 [Intermediate] • Target: Finding an unknown constant in a pdf

Find the constant term of the quotient when $4x^2 + 6x + 3$ is divided by $(x+1)$.

a. **2** ✓

b. 1

c. 4

d. 3

Dividing: quotient $= 4x + 2$, remainder $= 1$. Constant term of quotient: 2.

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32 [Intermediate] • Target: Finding probabilities by integrating the pdf

Consider the exponential growth model described by the function $P(t) = P_0e^{kt}$, where $P(t)$ is the population at time t, P_0 is the initial population, and k is the growth constant. If the initial population P_0 is 500 and the population after 4 years is 2000, determine the value of the growth constant k to three decimal places.

a. **0.347** ✓

b. 0.416

c. 0.301

d. 0.477

The population model is given by $P(t) = P_0e^{kt}$. We are given that $P_0 = 500$. We are also given that $P(4) = 2000$. Substituting these values into the equation, we get: $2000 = 500e^{k \cdot 4}$ Divide both sides by 500: $4 = e^{4k}$ Take the natural logarithm of both sides: $\ln(4) = \ln(e^{4k})$ $\ln(4) = 4k$ Solve for k: $k = \frac{\ln(4)}{4}$ Using a calculator, $\ln(4) \approx 1.38629$. $k \approx \frac{1.38629}{4}$ $k \approx 0.34657$ Rounding to three decimal places, $k \approx 0.347$.

33 [Intermediate] • Target: The cumulative distribution function

A random variable $X \sim B(100, 0.6)$ is approximated by a normal distribution. State the mean of the approximating normal distribution.

a. 58.0

b. 65.0

c. **60** ✓

d. 62.0

Applying Normal Distribution principles, Mean $= np = 100 \times 0.6 = 60$ Therefore, the verified correct option is 60.

34 [Intermediate] • Target: Using the cdf to find probabilities

A discrete random variable has $P(X = 1) = k$, $P(X = 2) = 2k$, $P(X = 3) = 3k$. Find k.

a. $\frac{5}{12}$

b. $\frac{2}{3}$

c. $\frac{1}{6}$ ✓

d. $\frac{5}{6}$

Working through this statistics step: $k + 2k + 3k = 1 \Rightarrow 6k = 1 \Rightarrow k = \frac{1}{6}$

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35 [Intermediate] • Target: Expectation of a continuous random variable

A random variable X equals 2 with probability $\frac{1}{5}$ and 4 with probability $\frac{4}{5}$. Find $\mathrm{Var}(X)$.

a. $\frac{2}{5}$

b. $\frac{16}{25}$ ✓

c. $\frac{33}{25}$

d. $\frac{39}{50}$

$E(X) = \frac{18}{5},\ E(X^2) = \frac{68}{5},\ \mathrm{Var}(X) = E(X^2) - [E(X)]^2 = \frac{16}{25}$

36 [Intermediate] • Target: Variance of a continuous random variable

A continuous function f and a constant a satisfy $\int_a^x f(t)\,dt = e^x - e^5$ for all x, with $a > 0$. Find a.

a. 25

b. -5

c. 10

d. 5 ✓

Setting $x = a$ makes the integral 0, so $e^a - e^5 = 0$, giving $a = 5$.

37 [Intermediate] • Target: Median and quartiles

A discrete random variable X is uniformly distributed on $\{1, 2, \ldots, 7\}$. Find $E(X)$.

a. 9/2

b. 3

c. **4** ✓

d. 5

$E(X) = \frac{n+1}{2} = \frac{7+1}{2} = 4$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: A well-designed word problem: waiting time at a bus stop • Cross-Lesson: 3.01 Advanced Trigonometry

The histogram below shows the distribution of heights of a group of students. The total number of students in the group is 50. If the frequency density for the class interval $60 \le h < 70$ is 1.2, find the value of k.

a. 10

b. **12** ✓

c. 15

d. 20

The area of each bar in a histogram represents the frequency of the corresponding class interval. The frequency density is given by $\frac{\text{frequency}}{\text{class width}}$. The class interval $60 \le h < 70$ has a class width of $70 - 60 = 10$. The frequency density for this interval is given as 1.2. Therefore, the frequency for this interval is frequency density $\times$ class width $= 1.2 \times 10 = 12$. The histogram shows that the frequency for the interval $60 \le h < 70$ is represented by k..

39 [Challenging] • Target: Verifying a valid probability density function • Cross-Lesson: 4.01 Differentiation

The complex number z satisfies the equation $|z - (6 + 8\mathrm{i})| = 4$. Given that z^* denotes the complex conjugate of z, determine the maximum possible value of $|z - \mathrm{i}z^*|$.

a. $8 + 2\sqrt{2}$ ✓

b. $8 - 2\sqrt{2}$

c. $4\sqrt{2} + 2$

d. $4 + \sqrt{2}$

Let $z = x + \mathrm{i}y$, where $x, y \in \mathbb{R}$. Then $z^* = x - \mathrm{i}y$. First, express $z - \mathrm{i}z^*$ in Cartesian components: $\mathrm{i}z^* = \mathrm{i}(x - \mathrm{i}y) = y + \mathrm{i}x$, $z - \mathrm{i}z^* = (x + \mathrm{i}y) - (y + \mathrm{i}x) = (x - y) + \mathrm{i}(y - x) = (x - y)(1 - \mathrm{i})$. Taking the modulus on both sides: $|z - \mathrm{i}z^*| = |x - y| \cdot |1 - \mathrm{i}| = \sqrt{1^2 + (-1)^2}\,|x - y| = \sqrt{2}\,|x - y|$. The given condition $|z - (6 + 8\mathrm{i})| = 4$ d.

40 [Challenging] • Target: Finding an unknown constant in a pdf • Cross-Lesson: 5.01 Integration

The figure shows the graph of a continuous function $f(x)$ on the interval $[a, d]$. The shaded region represents the definite integral $\int_a^c f(x)\,dx$. If the area of the shaded region is 12 square units and the area of the unshaded region between $x = c$ and $x = d$ is 5 square units, what is the value of $\int_a^d f(x)\,dx$?

a. **17** ✓

b. 7

c. 2.4

d. -7

The Fundamental Theorem of Calculus relates the definite integral of a function to its antiderivative. Geometrically, the definite integral $\int_a^b f(x)\,dx$ represents the net signed area between the curve $f(x)$ and the x-axis from $x = a$ to $x = b$. Areas above the x-axis are positive, and areas below the x-axis are negative. The problem states that the area of the shaded region, representing $\int_a^c f(x)\,dx$, is 12 square units. This implies that this region contributes positively to the net signe.

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41 [Challenging] • Target: Finding probabilities by integrating the pdf • Cross-Lesson: 6.01 Sequences & Series

Which of the following is the correct expression of $\frac{5x+1}{(x-1)(x+2)}$ in partial fractions?

a. $\frac{2}{x-1}+\frac{3}{x+2}$ ✓

b. $\frac{3}{x-1}+\frac{2}{x+2}$

c. $\frac{2}{x-1}-\frac{3}{x+2}$

d. $\frac{1}{x-1}+\frac{4}{x+2}$

To decompose the rational function into partial fractions, set up the identity with distinct linear factors: $\frac{5x+1}{(x-1)(x+2)} \equiv \frac{A}{x-1}+\frac{B}{x+2}$ Multiplying both sides by the common denominator $(x-1)(x+2)$: $5x+1 \equiv A(x+2)+B(x-1)$ To determine A, substitute $x=1$: $5(1)+1=A(1+2)+B(0) \Rightarrow 6=3A \Rightarrow A=2$ To determine B, substitute $x=-2$: $5(-2)+1=A(0)+B(-2-1) \Rightarrow -9=-3B \Rightarrow B=3$ Therefore, the partial.

42 [Challenging] • Target: The cumulative distribution function • Cross-Lesson: 7.01 Vectors

Find the exact real solution to the equation $2e^{2x}+5e^{x}-12=0$.

a. $x=\ln\left(\frac{3}{2}\right)$ ✓

b. $x=\ln 4$

c. $x=\frac{3}{2}$

d. $x=\ln\left(\frac{3}{2}\right)$ and $x=\ln 4$

We recognize that the equation is quadratic in terms of e^{x}: $2(e^{x})^{2}+5(e^{x})-12=0$. Let $u=e^{x}$. Then the equation becomes: $2u^{2}+5u-12=0$. Factoring the quadratic expression: $(2u-3)(u+4)=0$. This gives two possible values for u: $u=\frac{3}{2}$ or $u=-4$. Since $u=e^{x}$: 1. For $e^{x}=\frac{3}{2}$, taking the natural logarithm of both sides yields: $x=\ln\left(\frac{3}{2}\right)$. 2. For $e^{x}=-4$, since $e^{x}>0$ for all real numbers x, there are no real solu.

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43 [Challenging] • Target: Using the cdf to find probabilities • Cross-Lesson: 8.01 Algebra: Partial Fractions & Division

The complex number z satisfies the equation $z(2-\mathrm{i}) = 7+4\mathrm{i}$. Express z in the form $x+\mathrm{i}y$, where x and y are real.

a. $2+3\mathrm{i}$ ✓

b. $\frac{18}{5}+\frac{1}{5}\mathrm{i}$

c. $2-3\mathrm{i}$

d. $\frac{10}{3}+5\mathrm{i}$

To find z, rearrange the given equation: $z = \frac{7+4\mathrm{i}}{2-\mathrm{i}}$ Multiply the numerator and the denominator by the complex conjugate of the denominator, which is $2+\mathrm{i}$: $z = \frac{(7+4\mathrm{i})(2+\mathrm{i})}{(2-\mathrm{i})(2+\mathrm{i})}$ Expand the denominator: $(2-\mathrm{i})(2+\mathrm{i}) = 2^2 - \mathrm{i}^2 = 4-(-1) = 5$ Expand the numerator.

44 [Challenging] • Target: Expectation of a continuous random variable • Cross-Lesson: 9.01 Complex Numbers

Find the exact value of the definite integral $\int_0^{\ln 2} \mathrm{e}^{2x}\,\mathrm{d}x$.

a. $\frac{3}{2}$ ✓

b. 3

c. $\frac{1}{2}$

d. $\frac{7}{2}$

To evaluate the definite integral: 1. Find the antiderivative using the standard standard result $\int \mathrm{e}^{kx}\,\mathrm{d}x = \frac{1}{k}\mathrm{e}^{kx}+C$: $\int \mathrm{e}^{2x}\,\mathrm{d}x = \frac{1}{2}\mathrm{e}^{2x}$ 2. Apply the limits from $x=0$ to $x=\ln 2$: $\left[\frac{1}{2}\mathrm{e}^{2x}\right]_0^{\ln 2} = \frac{1}{2}\mathrm{e}^{2\ln 2} - \frac{1}{2}\mathrm{e}^{2(0)}$ 3. Simplify the exponential terms using the power law of logarithms, $2\ln 2 = \ln(2^2) = \ln 4$.

45 [Challenging] • Target: Variance of a continuous random variable • Cross-Lesson: 1.01 Functions: Modulus & Inverse

Formulate the null and alternative hypotheses for the following scenario: "A drug is claimed to reduce blood pressure by 10 units on average. A skeptic thinks the true reduction is less than 10."

a. $H_0 : \mu = 250,\ H_1 : \mu e 250$

b. $H_0 : \mu = 70,\ H_1 : \mu > 70$

c. $H_0 : \mu = 10,\ H_1 : \mu < 10$ ✓

d. $H_0 : \mu = 500,\ H_1 : \mu < 500$

Applying Hypothesis Testing principles, $H_0 : \mu = 10,\ H_1 : \mu < 10$. Therefore, the verified correct option is $H_0 : \mu = 10,\ H_1 : \mu < 10$.

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